TOPIC 1.3
Representing Motion

Required Course Content

LEARNING OBJECTIVE	ESSENTIAL KNOWLEDGE
1.3.A Describe the position, velocity, and acceleration of an object using representations of that object's motion.	**1.3.A.1** Motion can be represented by motion diagrams, figures, graphs, equations, and narrative descriptions.
	1.3.A.2 For constant acceleration, three kinematic equations can be used to describe instantaneous linear motion in one dimension: $v_x = v_{x0} + a_x t$ $x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$ $v_x^2 = v_{x0}^2 + 2a_x (x - x_0)$ *Note: The equations above are written to indicate motion in the x-direction, but these equations can be used in any single dimension as appropriate.*
	1.3.A.3 Near the surface of Earth, the vertical acceleration caused by the force of gravity is downward, constant, and has a measured value approximately equal to $a_g = g \approx 10 \text{ m/s}^2$.
	1.3.A.4 Graphs of position, velocity, and acceleration as functions of time can be used to find the relationships between those quantities.

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SUGGESTED SKILLS

1.C Create qualitative sketches of graphs that represent features of a model or the behavior of the physical system.

2.A Derive a symbolic expression from known quantities by selecting and following a logical mathematical pathway.

2.D Predict new values or factors of change of physical quantities using functional dependence between variables.

3.C Justify or support a claim using evidence from experimental data, physical representations, or physical principles or laws.

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LEARNING OBJECTIVE

1.3.A

Describe the position, velocity, and acceleration of an object using representations of that object's motion.

ESSENTIAL KNOWLEDGE

1.3.A.4.i

An object's instantaneous velocity is the rate of change of the object's position, which is equal to the slope of a line tangent to a point on a graph of the object's position as a function of time.

Relevant equation:

$$v_x = \frac{dx}{dt}$$

1.3.A.4.ii

An object's instantaneous acceleration is the rate of change of the object's velocity, which is equal to the slope of a line tangent to a point on a graph of the object's velocity as a function of time.

Relevant equation:

$$a_x = \frac{dv_x}{dt}$$

1.3.A.4.iii

The displacement of an object during a time interval is equal to the area under the curve of a graph of the object's velocity as a function of time (i.e., the area bounded by the function and the horizontal axis for the appropriate interval).

Relevant equation:

$$\Delta x = \int_{t_1}^{t_2} v_x(t)\ dt$$

1.3.A.4.iv

The change in velocity of an object during a time interval is equal to the area under the curve of a graph of the acceleration of the object as a function of time.

Relevant equation:

$$\Delta v_x = \int_{t_1}^{t_2} a_x(t)\ dt$$

BOUNDARY STATEMENT

AP Physics C: Mechanics and AP Physics C: Electricity and Magnetism expects that for all situations in which a numerical quantity is required for g, the value $g \approx 10\ \text{m/s}^2$ will be used. However, students will not be penalized for correctly using the more precise commonly accepted values of $g = 9.81\ \text{m/s}^2$ or $g = 9.8\ \text{m/s}^2$.

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